

PRANAV N J

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Education

Siddaganga Institute of Technology , BE in Information Science	Sept 2023 – July 2027
• CGPA: 9.36/10	
Excellent PU College , PCMB	May 2023
• 12th: 94.5%	
JVS Primary and High School	May 2021
• 10th: 95.84%	

Skills

Languages: C++, C, JavaScript, Python
Frameworks: React.js, Node.js, Express.js, FastAPI
Databases: MongoDB, PostgreSQL, MySQL
ML/Data Libraries: NumPy, Pandas, Scikit-learn
Tools: Git, GitHub, VS Code, Docker
CS Fundamentals: Data Structures, DBMS, OOP, OS

Projects

Phishing Detection System MERN, Python, ML, FastAPI, REST API	GitHub
- Built an ML-powered phishing detection microservices (React, FastAPI, Express, MongoDB, Web Extension) that classify malicious URLs with 93.89% accuracy and 29 ms prediction latency. - Eliminated HTTPS protocol bias via dataset augmentation (27K→55K URLs) after evaluating 4 ML algorithms, reducing false positive rate to 4.79% and preventing production failure. - Developed 59-feature Random Forest classifier with GridSearchCV hyperparameter tuning and StratifiedKFold 5-fold cross-validation, improving precision by 7.1% and achieving 98.3% AUC-ROC.	
Suraksha TypeScript, PostgreSQL, WebSocket, WebRTC, CoreNLP	GitHub
- AI-powered women's safety platform built in TypeScript (React + Node + PostgreSQL) delivering sub-2 sec multi-channel emergency alerts, AI voice distress detection, and encrypted real-time coordination for families. - Engineered AI emergency voice distress detection with a 3-stage pipeline (AssemblyAI → CoreNLP → Llama 2), reaching 96% accuracy and cutting false positives by 35% via weighted keywords plus rule-based fallback. - Optimized 24/7 real-time location tracking with 15s GPS WebSocket beacons and WebRTC live video streaming, sustaining 97.8% multi-channel delivery across 37+ active emergency sessions.	
Server-Monitor MySQL, Python, FastAPI, WebSocket, React, psutil	GitHub
- Built a database-centric server monitoring system using MySQL triggers and stored procedures for instant server health alerts, with a real-time WebSocket pipeline pushing live CPU/RAM/Disk metrics every 3 seconds. - Automated multi-channel SMS and email alerts for critical server issues with SQL-enforced 5-minute cooldown, centralized dispatch logging, and minimal backend complexity. - Engineered database-driven Recharts visualizations for metric history, uptime tracking, and paginated logs improving performance 75% across time ranges.	

Achievements

Technical Competitions: Selected participant for the **Smart India Hackathon 2025**, a premier national-level hackathon by the Government of India; Participant at **HackNite'25** organized by MIT Bengaluru; Participant at **National Space Hackathon** hosted by IIT Delhi.

Leadership and Service: Certified NCC Cadet who mentored 20–30 junior cadets in leadership training; Event Coordinator for **Institution's Innovation Council (IIC-SET)** organizing recruitment and innovation initiatives.